Mitoprep Premium Test 7

Date · 14 January 2026

Physics :- Thermal Properties of Matter, Thermodynamics, Kinetic Theory of Gases

Chemistry :- Organic Chemistry: Some Basic Principles & Techniques,
Hydrocarbons

Biology :- Photosynthesis in Higher Plants, Respiration in Plants,
Plant Growth & Development, Locomotion & Movement,
Neural Control & Coordination, Chemical Coordination & Integration

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

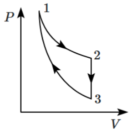
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. Three processes compose a thermodynamic cycle shown in the accompanying (P-V) diagram of an ideal gas.



Process $1 \rightarrow 2$ takes place at constant temperature (300 K). During this process 60 J of heat enters the system.
Process $2 \rightarrow 3$ takes place at constant volume. During this process 40 J of heat leaves the system.
Process $3 \rightarrow 1$ is adiabatic, where the temperature at state 3 is $T_3 = 275$ K.

What is the change in the internal energy of the system during process $3 \rightarrow 1$?
1. -40 J
2. -20 J
3. 0 J
4. 40 J

2. One mole of an equimolar mixture of monoatomic (He) and diatomic (H_2) gases is heated to raise the temperature by 1 K under constant pressure. The amount of heat used in this process is (nearly):
1. 8.3 J
2. 16.6 J
3. 25 J
4. 29 J

3. Different types of gas molecules are mentioned in **Column-I**, while their degrees of freedom are given in **Column-II**. Match them.

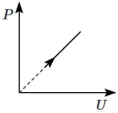
	Column-I	Column-II	
(A)	Monatomic gas	(I)	7
(B)	Diatomic gas (no vibration)	(II)	6
(C)	Diatomic gas (with vibration)	(III)	5
(D)	Triatomic gas (water) (no vibration)	(IV)	3

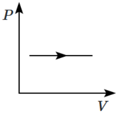
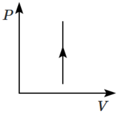
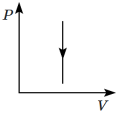
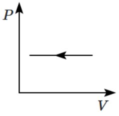
1.	A-IV, B-III, C-I, D-II
2.	A-IV, B-II, C-III, D-I
3.	A-IV, B-III, C-III, D-II
4.	A-IV, B-III, C-II, D-I

4. In the adiabatic compression, the decrease in volume is associated with:

1.	increase in temperature and decrease in pressure
2.	decrease in temperature and increase in pressure
3.	decrease in temperature and decrease in pressure
4.	increase in temperature and increase in pressure

5. The given $(P\text{-}U)$ graph shows how the internal energy of an ideal gas changes with increasing pressure. Which of the following pressure-volume $(P\text{-}V)$ graphs corresponds to this relationship?

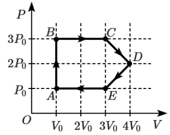


1.		2.	
3.		4.	

6. An object cools down from 80°C to 60°C in 5 min in a surrounding of temperature 20°C. The time taken to cool from 60°C to 40°C will be:
(Assume Newton's law of cooling to be valid.)

1.	$\frac{25}{3}$ min	2.	5 min
3.	$\frac{25}{4}$ min	4.	9 min

7. An ideal gas undergoes a cyclic process as shown in the pressure, P versus volume, V graph.



At which point does the gas reach its highest temperature?

1.	A	2.	B
3.	C	4.	D

8. What is the molar-specific heat capacity of a diatomic gas in an isochoric process if it has an additional vibrational mode?

1. $\frac{5}{2}R$
2. $\frac{3}{2}R$
3. $\frac{7}{2}R$
4. $\frac{9}{2}R$

9. The volume occupied by the molecules contained in 4.5 kg water at STP, if the molecular forces vanish away, is:
1. 5.6 m^3
2. 5.6×10^6 m^3
3. 5.6×10^3 m^3
4. 5.6×10^{-3} m^3

10. Given below are two statements:

Statement I:	If a mixture of ideal gases has specific molar heat capacities C_p (at constant pressure) and C_v (at constant volume) then, $C_p - C_v = R$.
Statement II:	Any mixture of ideal non-reacting gases will obey the ideal gas law: $pV = nRT$.

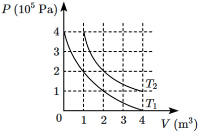
1.	**Statement I** is incorrect and **Statement II** is correct.
2.	Both **Statement I** and **Statement II** are correct.
3.	Both **Statement I** and **Statement II** are incorrect.
4.	**Statement I** is correct and **Statement II** is incorrect.

11. Physical quantities associated with an ideal gas undergoing a process are indicated in **Column-I** while their values are given in a different order in **Column-II**. Match them. R is the universal gas constant.

Column-I		**Column-II**	
(A)	Molar specific heat of a monatomic gas in an isobaric process (as a multiple of R)	(I)	1
(B)	Molar specific heat of a diatomic gas in an isochoric process (as a multiple of R)	(II)	$\frac{3}{2}$
(C)	Bulk modulus of elasticity as a multiple of pressure for a monatomic gas undergoing an adiabatic process	(III)	$\frac{5}{3}$
(D)	Bulk modulus of elasticity as a multiple of pressure for a diatomic gas undergoing an isothermal process	(IV)	$\frac{5}{2}$

1.	A-IV, B-IV, C-III, D-I
2.	A-II, B-IV, C-I, D-III
3.	A-IV, B-II, C-III, D-I
4.	A-II, B-II, C-I, D-III

12. The following graph shows two isotherms for a fixed mass of an ideal gas. The ratio of rms speed of the molecules at temperatures T_2 and T_1 is:

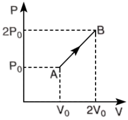


1.	$2\sqrt{2}$	2.	$\sqrt{2}$
3.	2	4.	4

13. The efficiency of a Carnot engine operating between two reservoirs is 25%. The ratio of the absolute temperature of the hot reservoir to that of the cold reservoir is:

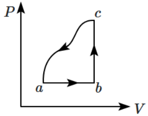
1.	4	2.	2
3.	$\frac{3}{2}$	4.	$\frac{4}{3}$

14. The P-V diagram of 2 g of helium gas for a certain process $A \rightarrow B$ is shown in the figure. What is the heat given to the gas during the process $A \rightarrow B$?



1.	$4P_0V_0$	2.	$6P_0V_0$
3.	$4.5P_0V_0$	4.	$2P_0V_0$

15. A sample of an ideal gas is taken through the cyclic process $abca$ as shown in the figure. The change in the internal energy of the gas along the path ca is -180 J. The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is:



1. 140 J
2. 130 J
3. 100 J
4. 120 J

16. If the same amount of heat is supplied to two spheres of the same material having the same radius (one is hollow and the other is solid), then:

1.	the expansion in hollow is greater than the expansion in solid
2.	the expansion in hollow is the same as that in solid
3.	the expansion in hollow is lesser than in solid
4.	the temperature of both must be the same

17. Two rods of equal length and area of the cross-section are kept parallel and lagged between temperatures $20^{\circ}\mathrm{C}$ and $80^{\circ}\mathrm{C}$. The ratio of the effective thermal conductivity to that of the first rod is:
(the ratio $\left(\frac{K_1}{K_2}=\frac{3}{4}\right)$)

1.	7 : 4	2.	7 : 6
3.	4 : 7	4.	7 : 8

18. On increasing the number density of a gas in a vessel, the mean free path of a gas:
1. decreases
2. increases
3. remains the same
4. becomes double

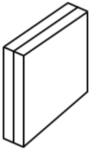
19. If the temperature (in $^{\circ}\mathrm{C}$) of a blackbody is increased 2-fold, then the rate of radiation from the body will become:

1.	16-fold	2.	4-fold
3.	less than 16-fold	4.	more than 16-fold

20. The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_P and C_V, respectively. If $\gamma=\frac{C_P}{C_V}$ and R is the universal gas constant, then C_V is equal to:

1.	$\dfrac{R}{\gamma-1}$	2.	$\dfrac{\gamma-1}{R}$
3.	γR	4.	$\dfrac{(\gamma-1)R}{(\gamma+1)}$

21. Two metal plates of area 80 cm^2 and thickness 3 mm, are soldered together (see figure). The face of the left plate is maintained at $100^{\circ}\mathrm{C}$, and the face of the right plate is maintained at $0^{\circ}\mathrm{C}$. The thermal conductivity of the left plate is 48 $\mathrm{Wm}^{-1}\,\mathrm{K}^{-1}$, while that of the right plate is 68 $\mathrm{Wm}^{-1}\,\mathrm{K}^{-1}$. What is the approximate temperature at the soldered junction?



1.	$21^{\circ}\mathrm{C}$	2.	$32^{\circ}\mathrm{C}$
3.	$41^{\circ}\mathrm{C}$	4.	$61^{\circ}\mathrm{C}$

22. A metal ball of mass 2 kg is heated by a 30 W heater, in a room at $20^{\circ}\mathrm{C}$. The temperature of the metal becomes steady at $50^{\circ}\mathrm{C}$. If the same ball was heated by a 20 W heater in a room at $30^{\circ}\mathrm{C}$, the steady temperature of the ball will be:

1.	$40^{\circ}\mathrm{C}$	2.	$50^{\circ}\mathrm{C}$
3.	$60^{\circ}\mathrm{C}$	4.	$70^{\circ}\mathrm{C}$

23. The molar specific heat (at constant volume) of a monoatomic ideal gas is $C_v = \dfrac{3}{2}R$ while that of a diatomic ideal gas is $C_v = \dfrac{5}{2}R$, where R is the universal gas constant. The molar specific heat (at constant pressure) of an equimolar mixture of the two will be:

1. $\dfrac{1}{2}\left(\dfrac{3}{2}R + \dfrac{5}{2}R\right)$
2. $\left(\dfrac{2}{3} + \dfrac{2}{5}\right)^{-1} R$
3. $3R$
4. $\dfrac{8}{15}R$

24. A piece of alloy of mass 250 g (specific heat capacity $= 0.1\times$ that of water) is placed in a furnace and then put into a calorimeter containing 240 g of water at 20°C. The water equivalent of the calorimeter is 10 g. The final temperature of the mixture is 50°C. The temperature of the furnace is (nearly):

1.	250°C	2.	350°C
3.	600°C	4.	800°C

25. The rise in the temperature of a given mass of an ideal gas at constant pressure and at an initial temperature 27°C to double its volume is:

1. 327°C
2. 54°C
3. 300°C
4. 600°C

26. Suppose that the RMS rotational kinetic energy of a diatomic molecule in a gas is written as:

$E_{\text{RMS}} = \dfrac{1}{2} I \omega_{\text{RMS}}^2$

where ω_{RMS} is the RMS angular speed, and I is the moment of inertia about the axis through centre-of-mass, perpendicular to the plane of rotation. The variation of ω_{RMS} with temperature is given by:

1. $\propto T^{1/2}$
2. $\propto T$
3. $\propto T^{-1/2}$
4. $\propto T^{-1}$

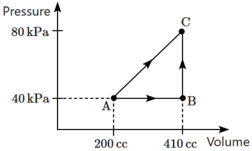
27. The average internal kinetic energy, due to translational motion of a molecule in an ideal gas at absolute temperature T, is E_{tr}. This quantity, E_{tr}, depends on:

(A)	mass of a molecule
(B)	number of atoms in a molecule
(C)	temperature, T

Choose the correct option from the given ones:

1.	**(C)** only	2.	**(B)** only
3.	**(B)**, **(C)** only	4.	**(A)**, **(B)**, **(C)**

28. The quantity of heat required to take a system from A to C through the process ABC is 20 cal. The quantity of heat required to go from A to C directly is:



1. 20 cal
2. 24.2 cal
3. 21 cal
4. 23 cal

29. Compute the ratio of specific heats (C_P/C_V) of an equimolar mixture of hydrogen and helium.

1.	$\dfrac{23}{15}$	2.	$\dfrac{3}{2}$
3.	2	4.	$\dfrac{7}{3}$

30. The pressure of an ideal gas $\left(\gamma = \dfrac{3}{2}\right)$ is increased by 1% in an adiabatic process. The temperature of the gas:

1.	increases by 1.5%
2.	decreases by 1.5%
3.	increases by $\frac{1}{3}\%$
4.	increases by $\frac{2}{3}\%$

31. The specific latent heat of ice is 80 cal/g (for melting) and that of water is 540 cal/g (for vaporisation). What quantity of ice would be needed to condense m grams of steam at 100°C and cool it to 0°C?

1. $\frac{27m}{4}$ grams
2. $\frac{59m}{13}$ grams
3. $8m$ grams
4. $9m$ grams

32. An ideal gas at absolute temperature T is contained in a cubical vessel of side L. The average momentum of the gas molecules, in a direction parallel to a side of the vessel, is:

1.	$\propto T$	2.	$\propto \sqrt{T}$
3.	$T^{-1/2}$	4.	zero

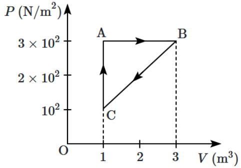
33. The equilibrium state of a thermodynamic system is described by:

A.	Pressure	**B.**	Total heat
C.	Temperature	**D.**	Volume
E.	Work done		

Choose the most appropriate answer from the options given below:

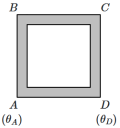
1.	**A**, **B** and **E** only	2.	**B**, **C** and **D** only
3.	**A**, **B** and **C** only	4.	**A**, **C** and **D** only

34. For the given cycle, the work done during the isobaric process is:



1. 200 J
2. zero
3. 400 J
4. 600 J

35. A square metallic frame is made from four identical rods connected end-to-end, as shown. The junctions A & D are maintained at temperatures θ_A, θ_D $(\theta_A > \theta_D)$. Heat can only flow along the length of the rods. The ratio of the thermal current in BC to that in AD is:

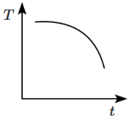
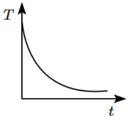
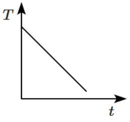
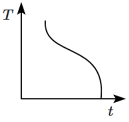


1.	1	2.	$\frac{1}{2}$
3.	$\frac{1}{3}$	4.	$\frac{1}{4}$

36. A slab of stone with an area $0.36\ \text{m}^2$ and thickness of 0.1 m is exposed on the lower surface to steam at 100° C. A block of ice at 0°C rests on the upper surface of the slab. In one hour 4.8 kg of ice is melted. The thermal conductivity of the slab will be:
(Given latent heat of fusion of ice $= 3.36 \times 10^5$ JKg^{-1})

1. 1.29 J/m/s/$^{\circ}$C
2. 2.05 J/m/s/$^{\circ}$C
3. 1.02 J/m/s/$^{\circ}$C
4. 1.24 J/m/s/$^{\circ}$C

37. A metal block is heated to a temperature much higher than room temperature and is allowed to cool in a room free from air currents. Which of the following curves correctly represents the cooling process?
(T represents the temperature of the block and t represents time)

1.		2.	
3.		4.	

38. A block of iron at 100°C is transferred to a plastic cup containing water at 20°C. Which one of the following is not necessary in order to find the specific heat capacity of iron?

1.	specific heat capacity of water
2.	mass of the block of iron
3.	thermal conductivity of the iron
4.	final temperature

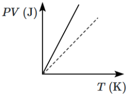
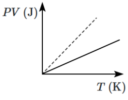
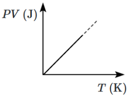
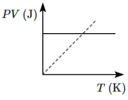
39. If the radius of a star is R and it acts as a black body, what would be the temperature of the star at which the rate of energy production is Q?
(σ is Stefan-Boltzmann constant)

1. $\dfrac{Q}{4\pi R^2\sigma}$
2. $\left(\dfrac{Q}{4\pi R^2\sigma}\right)^{\frac{-1}{2}}$
3. $\left(\dfrac{4\pi R^2 Q}{\sigma}\right)^{\frac{1}{4}}$
4. $\left(\dfrac{Q}{4\pi R^2\sigma}\right)^{\frac{1}{4}}$

40. Gas molecules at a temperature T are enclosed within a thin-walled enclosure, the inner and outer walls having the same temperature T in equilibrium. The rms speed of the gas molecules is v_{rms}. The enclosure radiates outward as a blackbody, and the most probable wavelength is λ_{mp}, in the blackbody radiation. As the temperature of the gas varies, so do v_{rms} and λ_{mp}. The relationship between v_{rms} and λ_{mp} is:

1.	$\lambda_{\text{mp}} v_{\text{rms}} = \text{constant}$	2.	$\dfrac{\lambda_{\text{mp}}}{v_{\text{rms}}} = \text{constant}$
3.	$\lambda_{\text{mp}} v_{\text{rms}}^2 = \text{constant}$	4.	$\dfrac{\lambda_{\text{mp}}}{v_{\text{rms}}^2} = \text{constant}$

41. Which one of the following schematic graphs best represents the variation of PV (in joules) versus T (in kelvin) of one mole of an ideal gas? (The dotted line represents $PV = T$).

1.		2.	
3.		4.	

42. An ice block (at 0°) is placed in the sun and it melts, producing a flow of water at the rate of $10\ \text{ml/s}$. The specific latent heat of fusion of ice is $80\ \text{cal g}^{-1}$. The quantity of thermal power absorbed by the ice is (nearly):
1. 3.36 kW
2. 336 kW
3. 0.8 kW
4. 800 kW

43. A monoatomic gas at a pressure P, having a volume V, expands isothermally to a volume $2V$ and then adiabatically to a volume $16V$. The final pressure of the gas is: $\left(\text{Take: } \gamma = \frac{5}{3}\right)$

1.	$64P$	2.	$32P$
3.	$\frac{P}{64}$	4.	$16P$

44. Which of the following relations does not give the equation of an adiabatic process, where terms have their usual meaning?

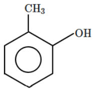
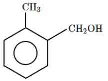
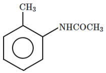
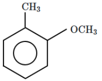
1. $P^{1-\gamma}T^{\gamma} = \text{constant}$
2. $PV^{\gamma} = \text{constant}$
3. $TV^{\gamma-1} = \text{constant}$
4. $P^{\gamma}T^{1-\gamma} = \text{constant}$

45. The pressure (p) of a mono-atomic gas can be related to the internal energy per unit volume (u) by the equation:

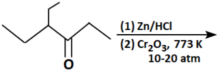
1.	$p = \frac{2}{3}u$	2.	$u = \frac{2}{3}p$
3.	$p = \frac{1}{3}u$	4.	$u = \frac{1}{3}p$

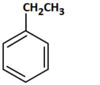
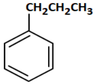
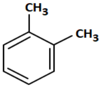
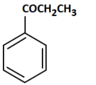
Chemistry

46. Which of the following is most reactive towards electrophilic reagents?

1.		2.	
3.		4.	

47. The major product among the following reaction is:

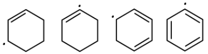


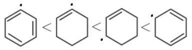
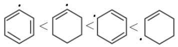
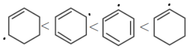
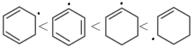
1.		2.	
3.		4.	

48. Which substrate is appropriate for the preparation of methane and ethane in a single-step process?

1. $H_2C = CH_2$
2. CH_3OH
3. CH_3-Br
4. CH_3-CH_2-OH

49. The increasing order of stability of the compounds given below is:



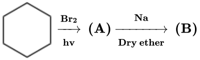
1.	
2.	
3.	
4.	

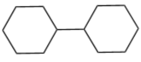
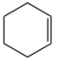
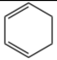
50. Which of the following statements is correct?

1.	In Kjeldahl's method, the compound is heated with copper oxide while in the Dumas method, compound is heated with concentrated sulphuric acid
2.	In the Dumas method, the compound is heated with ammonium while in Kjeldahl's method, compound is heated with concentrated sulphuric acid
3.	In Kjeldahl's method, the compound is heated with copper oxide while in the Dumas method, compound is heated with concentrated nitric acid
4.	In the Dumas method, the compound is heated with copper oxide while in Kjeldahl's method, compound is heated with concentrated sulphuric acid.

51. 0.40 g of an organic compound containing phosphorus gave 0.555g of $Mg_2P_2O_7$ by usual analysis. The % of phosphorus in the organic compound is:
1. 44.76
2. 38.75
3. 52.34
4. 66.44

52. Product B in the given reaction is:



1.		2.	
3.		4.	

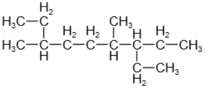
53. The principle involved in thin-layer chromatography is:
1. Partition
2. Sublimation
3. Adsorption
4. Solubility

54. What type(s) of chemical reaction is/are used to synthesize alkenes?
1. Substitution
2. Only elimination
3. Only addition
4. Addition as well as elimination reaction

55. Which of the following sodium compound(s) is/are formed when an organic compound containing both nitrogen and sulphur is fused with sodium?

1.	Cyanide and sulphide.	2.	Thiocyanate.
3.	Sulphite and cyanide.	4.	Nitrate and sulphide.

56. The correct IUPAC name for this non-polar alkane is:



1. 3-Ethyl-4-methylheptane
2. 3-Ethyl-4, 7-dimethynonane
3. 3-Methyl-7-ethydecane
4. 3,4-Diethyl-5, 7-dimethylnonane

57. What type of mechanism governs the chlorination reaction of methane?
1. Elimination
2. Substitution nucleophilic unimolecular (S_{N^1})
3. Free Radical
4. Substitution nucleophilic bimolecular (S_{N^2})

58. The total number of isomers for $C_4H_{10}O$ are (including stereoisomers):
1. 4
2. 5
3. 7
4. 8

59.

Assertion (A):	$CH_3 - C \equiv CH$ is more polar than CH_3 - CH = CH_2.
Reason (R):	sp hybridised carbon is more electro-negative than sp^2 hybridised carbon.

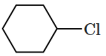
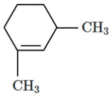
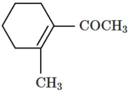
1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

60. The incorrect order regarding the – *I* effect of the substituents among the following is:

1.	$-\mathrm{I} < -\mathrm{Br} < -\mathrm{Cl} < -\mathrm{F}$
2.	$-\overset{+}{\mathrm{N}}\mathrm{R}_3 < -\overset{+}{\mathrm{O}}\mathrm{R}_2$
3.	$-\overset{+}{\mathrm{N}}\mathrm{R}_3 < -\mathrm{OR} < -\mathrm{F}$
4.	$-\mathrm{SR} < -\mathrm{OR} < -\overset{+}{\mathrm{O}}\mathrm{R}_2$

61. The technique used to purify liquids having high boiling points and getting decomposed at or below their boiling points is:
1. Fractional distillation
2. Steam distillation
3. Simple distillation
4. Distillation under reduced pressure

62. In which of the following molecules all the effects namely inductive, mesomeric and hyperconjugation operate?

1.		2.	
3.		4.	

63. 3-Methylpentane forms all the possible monohalogenated products. Among all monohalogenated products, how many products can show optical isomerism?
1. 1
2. 2
3. 3
4. 4

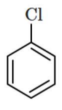
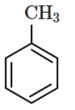
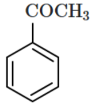
64. Based on the hyperconjugation effect described above,
identify the compound in which the double bond is the longest:
1. CH_3- CH = CH - CH_3
2. $(CH_3)_3$ C-CH = CH_2
3. CH_3-CH = CH-CH$(CH_3)_2$
4. CH_2 = CH- CH_2 -CH_3

65. In the following reaction, if 'X' is an alkyne with the lowest molecular weight, then the maximum number of atoms in 'Z' are:

$$X \xrightarrow[Fe\ or\ Cu]{Red\ Hot} \mathrm{Y} \xrightarrow[Anhyd.AlCl_3]{CH_3COCl} \mathrm{Z}$$

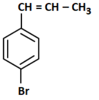
1. 15
2. 16
3. 14
4. 17

66. The increasing order of the reactivity of the following compounds toward electrophilic aromatic substitution reactions (EASR) is:

		
(I)	(II)	(III)

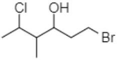
1. III < II < I
2. III < I < II
3. II < I < III
4. I < III < II

67. The reaction of $C_6H_5CH=CHCH_3$ with HBr produces:

1.	$C_6H_5\underset{\underset{Br}{\vert}}{C}HCH_2CH_3$
2.	$C_6H_5CH_2\underset{\underset{Br}{\vert}}{C}HCH_3$
3.	$C_6H_5CH_2CH_2CH_2Br$
4.	

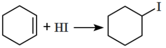
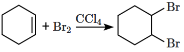
68. Which of the following compounds has the lowest boiling point?
1. $CH_3\ CH_2\ CH_2\ CH_2\ CH_3$
2. $CH_3\ CH = CH - CH_2\ CH_3$
3. $CH_3\ CH = CH - CH = CH_2$
4. $CH_3\ CH_2\ CH_2\ CH_3$

69. The correct IUPAC name of the following compound is :

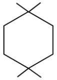
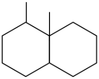
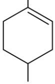
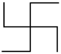


1. 6-Bromo-4-methyl-2-chlorohexan-4-ol
2. 1-Bromo-5-chloro-4-methylhexan-3-ol
3. 6-Bromo-2-chloro-4-methylhaxan-4-ol
4. 1-Bromo-4-methyl-5-chorohexan-3-ol

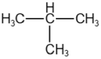
70. Which of the following reactions gives a wrong product under the given reaction conditions?

1.	
2.	$CH_3-(CH_2)_2-NH_2 \xrightarrow[HX]{NaNO_2} CH_3(CH_2)_2-NO_2+N_2$
3.	$CH_3-CH=CH_2+HBr \longrightarrow CH_3-\underset{\underset{Br}{\vert}}{C}H-CH_3$
4.	

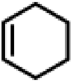
71. Which of the compound contains only $1^\circ, 2^\circ$ and 4° carbon atoms?

a.		b.	
c.		d.	

Choose the correct option:
1. (a), (b), (d)
2. (c), (d)
3. (a), (d)
4. (a), (b)

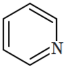
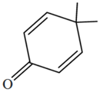
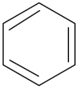
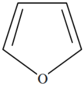
72. Which of the following pairs represent isomers?
1. $CH_3–CH_3$, $CH_3–CH_2–CH_3$
2. $H_2C=CH–OH$, $CH_3–O–CH_3$
3. $CH_3–CH_2–CH_2-CH_3$, 
4. All of the above.

73. The given compound is classified as:



1. Alicyclic
2. Aromatic
3. Antiaromatic
4. Acyclic

74. The least resonance stabilized molecule among the following is:

1.		2.	
3.		4.	

75. For an electrophilic substitution reaction, the presence of a halogen atom in the benzene ring:

A.	Deactivates the ring by inductive effect
B.	Deactivates the ring by resonance
C.	Increases the charge density at ortho and para position relative to meta position by resonance.
D.	Directs the incoming electrophile to meta position by increasing the charge density relative to ortho and para position.

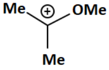
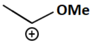
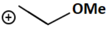
Choose the correct option
1. (A, B)
2. (B, C)
3. (C, D)
4. (A, C)

76. Consider the following reactions.

(i)	$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} A$ (major)
(ii)	$(CH_3)_3CH \xrightarrow{KMnO_4} B$ (major)

Major products (A) and (B), respectively, are:
1. Methanol and acetone
2. Methanal and Acetaldehyde
3. Methanal and 2-Methylpropan-2-ol
4. Methanol and 2-Methylpropan-2-ol

77. Find out the stability order of the intermediate:

(I)		(II)	
(III)		(IV)	$\overset{\oplus}{H_2C}$ – OMe

1. I > II > III > IV
2. I > II > IV > III
3. IV > I > II > III
4. III > II > I > IV

78. The substance that can be used as an adsorbent in thin-layer chromatography is:

1.	Na_2O	2.	Na_2SO_4
3.	Al_2O_3	4.	NaCl

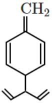
79. A hydrocarbon of molecular formula C_6H_{10} reacts with sodamide and the same on ozonolysis followed by hydrogen peroxide oxidation gives two molecules of carboxylic acids, one being optically active. The hydrocarbon maybe:

1.	1-Hexyne	2.	3-Hexyne
3.	3-Methyl-1-pentyne	4.	3,3-Dimethyl-1-butyne

80. The dihedral angle of the least stable conformer of ethane is:

1.	60°	2.	0°
3.	120°	4.	180°

81. The number of moles of hydrogen will be required for the complete hydrogenation of one mole of the following compound is:



1.	6	2.	7
3.	5	4.	3

82.

A.	$(C_6H_5)_3\dot{C}$
B.	$(CH_3)_2\dot{C}H$
C.	$(C_6H_5)_2\dot{C}\,CH_3$
D.	$C_6H_5\dot{C}(CH_3)_2$

For the free radical given above, correct option for decreasing order of stability is:
1. A > C > D > B
2. A > B > C > D
3. B > A > C > D
4. B > D > A > C

83. Match elements given in Column I with final product obtained during their qualitative analysis in column II and mark the correction option:

	Column I		Column II
(A)	Nitrogen	(P)	AgX
(B)	Sulphur	(Q)	$(NH_4)_3PO_4.12MoO_3$
(C)	Phosphorous	(R)	$Fe(SCN)_3$
(D)	Halogens	(S)	$Fe_4\left[Fe(CN)_6\right]_3$

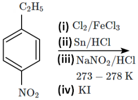
1. A(P), B(R), C(Q), D(S)
2. A(Q), B(R), C(Q), D(P)
3. A(S), B(R), C(Q), D(P)
4. A(Q), B(R), C(P), D(S)

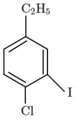
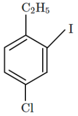
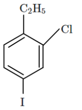
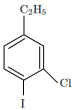
84. The effect of branching on an alkane is:
1. It increases its boiling point
2. It decreases its boiling point
3. It doesn't change its boiling point
4. None of the above

85. The correct number of σ and π bonds in the molecule CH_3NO_2 is:

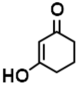
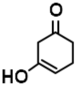
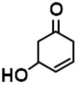
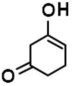
	σ (C - H)	σ (C - N)	σ (N - O)	π (N=O)
1.	1	1	1	3
2.	1	1	3	1
3.	1	3	1	3
4.	3	1	1	1

86. The major product X formed in the following reaction sequence is:



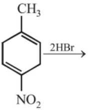
1.		2.	
3.		4.	

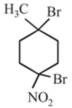
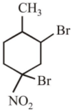
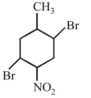
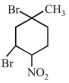
87. Compound, among the following, that cannot be a tautomer of 1,3-cyclohexanedione is:

1.		2.	
3.		4.	

88. How many optical isomers are formed by the hydrogenation of the compound $(CH_3)_2C = CHCH_3$?
1. 0
2. 1
3. 2
4. 3

89. The major product of the following reaction is :



1.		2.	
3.		4.	

90. The method used for quantitative estimation of halogens is:
1. Dumas method
2. Carius method
3. Kjeldahl's method
4. Combustion method

Biology

91. In a resting axonal membrane:

1.	The outer surface possesses a positive charge while its inner surface becomes negatively charged.
2.	The outer surface possesses a negative charge while its inner surface becomes positively charged.
3.	Both the outer and the inner surface are positively charged.
4.	Both the outer and the inner surface are negatively charged.

92. If metabolism consists solely of lipids, the respiratory quotient is approximately:

1.	0.7	2.	0.8
3.	0.9	4.	1.0

93. Consider the given two statements:

Statement I:	ATP synthase enzyme is located in the inner mitochondrial membrane.
Statement II:	The proton gradient is utilized by ATP synthase to synthesize ATP.

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is correct; **Statement II** is incorrect
3.	**Statement I** is incorrect; **Statement II** is correct
4.	**Statement I** is incorrect; **Statement II** is incorrect

94. Consider the two statements:

Statement I:	The electrons that are removed from photosystem II are replaced by electrons available due to splitting of water.
Statement II:	The electrons needed to replace those removed from photosystem I are provided by photosystem II.

1.	**Statement I** is correct; **Statement II** is incorrect
2.	**Statement I** is correct; **Statement II** is correct
3.	**Statement I** is incorrect; **Statement II** is incorrect
4.	**Statement I** is incorrect; **Statement II** is correct

95. Neurohypophysis:
1. secretes large number of tropins
2. synthesizes oxytocin and vasopressin
3. merges with pars distalis in humans
4. stores and releases oxytocin and vasopressin

96. Consider the given two statements:

Statement I:	In geometric growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures.
Statement II:	In arithmetic growth, both the progeny cells following mitotic cell division retain the ability to divide and continue to do so.

1.**Statement I** is correct; **Statement II** is incorrect
2. **Statement I** is correct; **Statement II** is correct
3. **Statement I** is incorrect; **Statement II** is incorrect
4.**Statement I** is incorrect; **Statement II** is correct

97. How many of the given statements are correct?

I:	The left and right cerebral hemispheres are connected by a tract of nerve fibres called corpus spongiosum.
II:	Association areas are neither clearly sensory nor motor in function.
III:	Thalamus is a major coordinating centre for sensory and motor signaling.
IV:	The hindbrain comprises pons, cerebellum and medulla.
V:	Three major regions make up the brain stem: mid brain, pons and medulla oblongata.

1.	2	2.	3
3.	4	4.	5

98. In the mitochondrial electron transport chain, Complex IV refers to:
1. NADH dehydrogenase
2. Cytochrome c oxidase complex
3. Ubiquinone
4. ATP synthase

99. Consider the given statements:

I:	The human skeletal system consists of 206 bones.
II:	The femur is the longest bone in the human body.
III:	Cartilaginous joints are immovable.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

100. Accessory pigments of photosynthesis in higher plants:

I:	include chlorophyll b, xanthophylls and carotenoids.
II:	absorb light and transfer electrons to chlorophyll a.
III:	protect chlorophyll a from photo-oxidation.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

101. Enzyme that catalyses the decarboxylation of pyruvic acid to acetaldehyde is:
1. Pyruvate dehydrogenase
2. Pyruvate kinase
3. Pyruvate decarboxylase
4. Pyruvate translocase

102. Consider the given statements:

I:	Wheat, rice and oat plants represent C_3 plants.
II:	Sugarcane and corn represent C_4 plants.
III:	Cacti and many desert plants are CAM plants.

1.	Only **I** and **II** are correct
2.	Only **I** and **III** are correct
3.	Only **II** and **III** are correct
4.	**I**, **II** and **III** are correct

103. What leads to the unmasking of active sites on actin filament for myosin filaments during muscle contraction?

1.	Release of acetylcholine by the anterior motor neuron at the neuromuscular junction.
2.	Binding of calcium with a subunit of troponin
3.	Binding of ATP on the myosin head
4.	Binding of acetylcholine to its receptor on the sarcolemma

104. Which of the following is a correct statement about the structure of a multipolar neuron?

1.	Neurons contain a single axon and several dendrites.
2.	Neurons do not have a nucleus in the cell body.
3.	The axon transmits electrical signals towards the cell body.
4.	Dendrites do not play a role in signal reception.

105. Given below are two statements:

Assertion (A):	Fermentation takes place under anaerobic conditions in prokaryotes only.
Reason (R):	In oxidative phosphorylation, water is the ultimate acceptor of electrons.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

106. Auxin was isolated from tips of coleoptiles of oat seedlings by:
1. Charles Darwin
2. F. W. Went
3. Boysen and Jensen
4. Francis Darwin

107. The correct chronological order [superficial to deep] of the meninges is:
1. dura mater - pia mater - arachnoid mater
2. dura mater - arachnoid mater - pia mater
3. pia mater - dura mater - arachnoid mater
4. pia mater - arachnoid mater - dura mater

108. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I		**Column II**
A	Chlorophyll *a*	P	Bright or blue green in chromatogram
B	Chlorophyll *b*	Q	Yellow green in chromatogram
C	PS I reaction centre	R	P680
D	PS II reaction centre	S	P700

Codes:

	A	**B**	**C**	**D**
1.	Q	P	R	S
2.	Q	P	S	R
3.	P	Q	S	R
4.	P	Q	R	S

109. Match the types of neurons in **Column-I** with their functions in **Column-II**:

	Column-I (Neuron Type)		Column-II (Function)
A.	Sensory Neuron	P.	Embryonic stages
B.	Motor Neuron	Q.	Carries impulses to muscles or glands
C.	Unipolar neuron	R.	Detects and transmits sensory stimuli
D.	Bipolar neuron	S.	Human retina

Options:
1. A-Q, B-P, C-S, D-R
2. A-S, B-Q, C-P, D-R
3. A-P, B-R, C-Q, D-S
4. A-R, B-Q, C-P, D-S

110. Consider the given two statements:

Assertion (A):	Unlike animals, plants retain the capacity for unlimited growth throughout their life.
Reason (R):	Unlike animals, there is presence of meristems at certain locations in their body.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	**(A)** is True; **(R)** is False
3.	**(A)** is False; **(R)** is False
4.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.

111. Match each item in **Column-I** with one item in **Column-II** and select the best match from the codes given:

	Column-I		Column-II
A.	H Zone	P.	Central Part of A Band
B.	A Band	Q.	Contains Myosin
C.	I Band	R.	Middle of H Zone
D.	M Line	S.	Contains Actin

Codes:

	A	B	C	D
1.	P	Q	R	S
2.	Q	P	R	S
3.	Q	P	S	R
4.	P	Q	S	R

112. There can be a net gain of ____ ATP molecules during aerobic respiration of one molecule of glucose [as per NCERT textbook]:
1. 30
2. 36
3. 38
4. 40

113. In an electrical synapse, how is the nerve impulse transmitted between neurons?

1.	By releasing neurotransmitters across the synaptic cleft
2.	Through the direct movement of ions via gap junctions
3.	By synaptic vesicles fusing with the postsynaptic membrane
4.	Through saltatory conduction at the nodes of Ranvier.

114. The anterior pituitary, also called the pars distalis, secretes all of the following hormones except:
1. Oxytocin
2. Growth hormone (GH)
3. Thyroid stimulating hormone (TSH)
4. Follicle stimulating hormone (FSH)

115. The axoplasm inside the axon contains:

1.	low concentration of K^+
2.	high concentration of negatively charged proteins
3.	high concentration of Na^+
4.	equal concentrations of Na^+ and K^+

116. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I [Region of adrenal cortex]		**Column I Hormone secreted**
A	zona reticularis	P	Aldosterone
B	zona fasciculata	Q	Cortisol
C	zona glomerulosa	R	Androgens

Codes:

	A	**B**	**C**
1.	P	Q	R
2.	R	Q	P
3.	Q	P	R
4.	P	R	Q

117. T.W Engelmann:

I:	described first absorption spectrum of photosynthesis.
ii:	first showed that it is only the green part of the plants that could release oxygen.

1.	Only **I** is correct
2.	Only **II** is correct
3.	Both **I** and **II** are correct
4.	Both **I** and **II** are incorrect

118. As the distance from Z line to Z line decreases, which bands shorten in a sarcomere?

1.	the I band and H band	2.	the I band and A band
3.	the A band and H band	4.	all bands shorten

119. Osteoporosis, a condition characterized by decreased bone mass and increased fragility, is primarily caused by the deficiency of:
1. Iron
2. Magnesium
3. Oestrogen
4. Progesterone

120. PTH is a hypercalcemic hormone. Which of the following statements will explain this function of PTH?

I:	PTH acts on bones and stimulates the process of bone resorption (dissolution/demineralisation).
II:	PTH stimulates reabsorption of Ca^{2+} by the renal tubules
III:	PTH increases Ca^{2+} absorption from the digested food

1.	Only **I** and **II**	2.	Only **I** and **III**
3.	Only **II** and **III**	4.	**I**, **II** and **III**

121. During oxygenic photosynthesis in higher plants, in the light reactions, the source of replacement electrons for Photosystem II is:
1. Glucose
2. Water
3. NADPH
4. Photosystem I

122. The coxal bone in the human skeleton:

1.	is a part of the pectoral girdle, has a glenoid cavity that articulates with the head of the femur.
2.	is a part of the pelvic girdle, has an acetabulum that articulates with the head of the humerus.
3.	is a part of the pelvic girdle and is formed by the fusion of three bones.
4.	is a part of pectoral girdle and is formed by the fusion of three bones.

123. Tetany is caused by:

1.	Hypocalcaemia leading to excessive neuronal excitability.
2.	Hyperkalaemia affecting muscle function.
3.	Deficiency of myosin ATPase activity.
4.	Increased sodium absorption in muscles.

124. Match the parts of the brain with their functions:

	Column I		**Column II**
A.	Cerebrum	i.	Coordination of movement and balance
B.	Cerebellum	ii.	Regulation of heart rate and breathing
C.	Medulla oblongata	iii.	Seat of intelligence, memory, and emotions

1. A-iii, B-i, C-ii
2. A-ii, B-iii, C-i
3. A-i, B-ii, C-iii
4. A-iii, B-ii, C-i

125. Consider the given two statements:

Assertion (A):	Growth, at the cellular level, is measured by a variety of parameters, some of which are: increase in fresh weight, dry weight, length, area, volume and cell number.
Reason (R):	Growth, at a cellular level, is principally a consequence of an increase in the amount of protoplasm, and increase in protoplasm is difficult to measure directly.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.
3.	**(A)** is True; **(R)** is False.
4.	**(A)** is False; **(R)** is True.

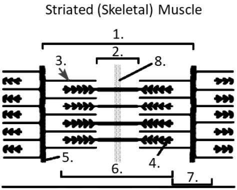
126. The primary pigment molecule responsible for capturing light energy during photosynthesis in higher plants is:
1. Chlorophyll
2. Carotenoid
3. Bacteriochlorophyll
4. Xanthophyll

127. How does an increase in CO_2 concentration generally affect the rate of photosynthesis in C_3 plants?

1.	Decreases the rate of photosynthesis
2.	Increases the rate of photosynthesis up to a point of saturation
3.	Has no effect on the rate of photosynthesis
4.	Causes photorespiration to increase

128. The number of carbon atoms are 3 in all of the following molecules except:
1. Dihydroxyacetone phosphate
2. Glyceraldehyde 3-phosphate
3. Acetyl CoA
4. Pyruvic acid

129. The length of which of the following regions shown in the given diagram of a sarcomere remains unchanged during muscle contraction?



1.	1	2.	2
3.	7	4.	6

130. The cells of which zone at the root apex attain their maximal size in terms of wall thickening and protoplasmic modifications?
1. Root cap
2. Meristematic zone
3. Elongation zone
4. Maturation zone

131. Plants can get along without respiratory organs. The reasons for this will include all the following except:

1.	Each plant part takes care of its own gas-exchange needs.
2.	Plants present great demands for gas exchange.
3.	The distance that gases must diffuse even in large, bulky plants is not great.
4.	Each living cell in a plant is located quite close to the surface of the plant.

132. Consider the given statements regarding the human vertebral column:

I:	In adults, it is differentiated into cervical [7], thoracic [12], lumbar [5], sacral [1-fused] and coccygeal [1-fused] regions.
II:	The first vertebra is called as axis and it articulates with the occipital condyles.
III:	The spinal cord runs through a neural canal.
IV:	It protects the spinal cord, supports the head and serves as the point of attachment for the ribs and musculature of the back.

The correct statements include:

1.	Only **I**	2.	Only **I** and **IV**
3.	Only **II** and **III**	4.	Only **I**, **III**, and **IV**

133. Aldosterone:

1.	acts mainly at the renal tubules and stimulates the excretion of Na^+ and water and reabsorption of K^+ and phosphate ions.
2.	acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions.
3.	acts mainly at the collecting duct and stimulates the excretion of Na^+ and water and reabsorption of K^+ and phosphate ions.
4.	acts mainly at the collecting duct and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions.

134. Addison's disease is:

1.	hypothyroidism	2.	hyperthyroidism
3.	hypoadrenalism	4.	hypopituitarism

135. Consider the given two statements:

Statement I:	Development in plants is the sum of two processes: growth and differentiation.
Statement II:	The development of a mature plant from a zygote (fertilised egg) follows a precise and highly ordered succession of events.

1. **Statement I** is correct; **Statement II** is incorrect
2. **Statement I** is correct; **Statement II** is correct
3. **Statement I** is incorrect; **Statement II** is incorrect
4. **Statement I** is incorrect; **Statement II** is correct

136. Which muscle type is more likely to contain abundant mitochondria and rely on aerobic processes for energy?
1. White muscle fibres
2. Red muscle fibres
3. Smooth muscle fibres
4. Cardiac muscle fibres

137. The sympathetic neural system and parasympathetic neural system are the subdivisions of:
1. Central neural system
2. Somatic neural system
3. Visceral neural system
4. Autonomic neural system

138. Consider the given statements:

I:	P680 is the reaction center of Photosystem II.
II:	The Z-scheme of electron transport occurs in the thylakoid membrane.
III:	The electron transport chain generates a proton gradient across the thylakoid membrane.

1.	Only **I** and **II** are correct
2.	Only **I** and **III** are correct
3.	Only **II** and **III** are correct
4.	**I**, **II** and **III** are correct

139. Consider the given two statements:

Assertion (A):	Water is essential for the growth of plants
Reason (R):	Water pressure (turgor pressure) within cells drives cell expansion and growth.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

140. How many of the given statements are correct?

I:	In males, LH stimulates the synthesis and secretion of hormones called androgens from testis.
II:	In males, FSH and androgens regulate spermatogenesis.
III:	In females, LH induces ovulation of fully mature follicles (graafian follicles) and maintains the corpus luteum, formed from the remnants of the graafian follicles after ovulation.
IV:	FSH stimulates growth and development of the ovarian follicles in females.

1.	1	2.	2
3.	3	4.	4

141. Which type of neuron, with one axon and one dendrite, is primarily found in the retina of the eye?
1. Unipolar
2. Multipolar
3. Bipolar
4. Quadripolar

142. During the conversion of succinyl-CoA to succinic acid in the TCA cycle, which molecule is directly synthesised?
1. GTP
2. ATP
3. NADH
4. $FADH_2$

143. Identify the incorrect statement:

1.	When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs due to cyclic flow of electrons.
2.	A possible location of cyclic flow of electrons is in the stroma lamellae.
3.	The stroma lamellae membranes have both PS II as well as NADP reductase enzyme.
4.	When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs due to cyclic flow of electrons.

144. 'The effect of water as a factor affecting photosynthesis is more through its effect on the plant, rather than directly on photosynthesis'. Which of the following support this statement?

I:	Water stress causes the stomata to close, hence reducing the CO_2 availability.
II:	Water stress also makes leaves wilt, thus, increasing the surface area of the leaves and their metabolic activity as well.

1.	Only **I**
2.	Only **II**
3.	Both **I** and **II**
4.	Neither **I** nor **II**

145. A patient suffering from hyperparathyroidism may have:
I: increased bone deposition
II: increased risk of fractures

1.	Only **I**	2.	Only **II**
3.	Both **I** and **II**	4.	Neither **I** nor **II**

146. The depolarization and repolarization phases of an action potential are respectively a result of:
1. Influx of Sodium; Influx of Potassium
2. Efflux of Sodium; Efflux of Potassium
3. Efflux of Sodium; Influx of Potassium
4. Influx of Sodium; Efflux of Potassium

147. In females, which hormone stimulates ovulation and maintains the corpus luteum?
1. Follicle stimulating hormone (FSH)
2. Prolactin
3. Luteinizing hormone (LH)
4. Growth hormone (GH)

148. Diabetes insipidus is caused due to:
1. an excessive secretion of insulin
2. a decreased secretion of insulin
3. an excessive secretion of ADH
4. a decreased secretion of ADH

149. Which one of the following pairs of hormones are the examples of those that can easily pass through the cell membrane of the target cell and bind to a receptor inside it (mostly in the nucleus)?
1. Insulin and glucagon
2. Thyroxin and insulin
3. Somatosatin and oxytocin
4. Cortisol and testosterone

150. During stress or emergency situations, the adrenal medulla releases catecholamines. Which of the following is not an effect of adrenaline/noradrenaline?

1.	Increased heart rate and cardiac contraction strength
2.	Vasodilation of blood vessels supplying skeletal muscles
3.	Decreased glycogen breakdown in liver cells
4.	Increased respiratory rate and alertness

151. How many of the given statements are correct:

I:	Gastrin acts on the gastric glands and inhibits the secretion of hydrochloric acid and pepsinogen.
II:	Secretin acts on the endocrine pancreas and inhibits secretion of water and bicarbonate ions.
III:	CCK acts on both pancreas and gall bladder and inhibits the secretion of pancreatic enzymes and bile juice, respectively.
IV:	GIP inhibits gastric secretion and motility.

1.	1	2.	2
3.	3	4.	4

152. The type of synovial joint between the first and the second cervical vertebra is called:

1.	Pivot	2.	Saddle
3.	Hinge	4.	Gliding

153. Which of the following has an ATPase activity?
1. Myosin head
2. Myosin tail
3. Troponin T
4. Tropomyosin

154. Photosynthetic Active Radiation (PAR) has the following range of wavelengths
1. 340-450 nm
2. 450-950 nm
3. 500-600 nm
4. 400-700 nm

155. Posterior pituitary gland does not secrete:
1. Oxytocin
2. Vasopressin (ADH)
3. Growth hormone
4. None of the above

156. Quantitative comparisons between the growth of living system can be made in two ways where:

I:	measurement and the comparison of total growth per unit time is called the absolute growth rate.
II:	the growth of the given system per unit time expressed on a common basis, e.g., per unit initial parameter is called the relative growth rate.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

157. Mitochondria are often referred to as the powerhouses of the cell due to their role in ATP production through oxidative phosphorylation. What structural feature of mitochondria is crucial for maximizing the surface area available for the electron transport chain?

1.	Cristae	2.	Matrix
3.	Outer membrane	4.	Intermembrane space

158. The ATP synthase complex in the ETS includes two main components, F_1 and F_0. The F_0 component's primary function is to:

1.	Bind ADP and inorganic phosphate to form ATP.
2.	Accept electrons directly from NADH.
3.	Transport ATP molecules out of the mitochondria.
4.	Channel protons from the intermembrane space to the matrix, facilitating ATP synthesis.

159. What happens during non-cyclic photophosphorylation?

1.	Only NADPH is made.
2.	Only ATP is made.
3.	Glucose is produced directly.
4.	Oxygen, ATP, and NADPH are formed.

160. How many of the following can be characterized as action of gibberellins on plants?

I:	They can be used to increase the length of grape stalks.
II:	They can elongate and improve the shape of apple.
III:	They speed up malting process in brewing industry.
IV:	They hasten the maturity period of juvenile conifers.
V:	They promote bolting in many plants with rosette habit.

1.	2	2.	3
3.	4	4.	5

161. How many of the following statements regarding plant growth regulators (PGRs) are correct?

I:	Auxins promote cell elongation and root growth in plants.
II:	Gibberellins are responsible for seed germination and stem elongation.
III:	Cytokinins inhibit apical dominance and promote lateral bud growth.
IV:	Ethylene stimulates fruit ripening and senescence in plants.

1. One
2. Two
3. Three
4. Four

162. Which of the following correctly represents the three main stages of the Calvin cycle in the correct order?

1.	Reduction, Carboxylation, Regeneration
2.	Carboxylation, Reduction, Regeneration
3.	Regeneration, Carboxylation, Reduction
4.	Reduction, Regeneration, Carboxylation

163. Consider the given two statements:

Assertion (A):	The protons and O_2 formed by photolysis of water are released in the thylakoid lumen.
Reason (R):	The water splitting complex is associated with the PS I, which itself is physically located on the outer side of the membrane of the thylakoid.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

164. Association areas in the cerebral cortex:

I:	are neither clearly sensory nor motor in function.
II:	are for complex functions like inter-sensory associations, memory and communication.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

165. In both lactate fermentation and alcohol fermentation:

1.	glucose is completely oxidized under aerobic conditions
2.	carbon dioxide is produced
3.	about 70 percent of energy trapped in glucose is released
4.	$NADH + H^+$ is reoxidized to NAD^+

166. Formation of interfascicular cambium and cork cambium in certain plants is an example of:

1.	Differentiation	2.	De-differentiation
3.	Re-differentiation	4.	Anti-differentiation

167. Select the correct statement with respect to locomotion in humans:

1.	Accumulation of uric crystals in joints causes inflammation
2.	The vertebral column has 10 thoracic vertebrae
3.	The joint between adjacent vertebrae is a fibrous joint
4.	The decreased level of progesterone causes osteoporosis in old people

168. Identify the incorrectly matched pair:

1.	Acromegaly	Excess of growth hormone in adulthood
2.	Diabetes insipidus	Deficiency of insulin
3.	Graves' disease	Hyperthyroidism
4.	Addison's disease	Underproduction of adrenal cortex hormones

169. Medulla oblongata:

I:	is a part of the brain stem
II:	is a part of the hind brain
III:	is a part of the limbic system

1.	Only **I** and **II**	2.	Only **I** and **III**
3.	Only **II** and **III**	4.	**I**, **II** and **III**

170. Which plant growth regulator is used to initiate flowering and synchronising fruit-set in pineapples?

1.	Gibberellins	2.	Cytokinins
3.	ABA	4.	Ethylene

171. The total number of floating ribs in human skeleton is:

1.	2	2.	4
3.	6	4.	14

172. Match the components of a sarcomere in **Column-I** with their roles in **Column-II**:

	Column-I (Component)		Column-II (Role)
A.	Actin Filament	P.	Forms the H-zone
B.	Myosin Filament	Q.	Thin filament with binding sites
C.	Z-line	R.	Marks the boundaries of a sarcomere
D.	A-band	S.	Overlapping region of thick and thin filaments

Options:
1. A-Q, B-P, C-R, D-S
2. A-P, B-S, C-Q, D-R
3. A-S, B-Q, C-P, D-R
4. A-Q, B-R, C-S, D-P

173. Consider the given two statements:

I:	When a neuron is not conducting any impulse, i.e., resting, the axonal membrane is comparatively more permeable to potassium ions ($K+$) and nearly impermeable to sodium ions ($Na+$).
II:	Similarly, the membrane is permeable to negatively charged proteins present in the axoplasm.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

174. Which plant growth regulator, also known as the stress hormone, stimulates the closure of stomata and increases the tolerance of plants to various kinds of stress?

1.	Ethylene	2.	Abscisic acid
3.	Gibberellin	4.	Auxin

175. A plant growth regulator can be sprayed on sugarcane to increase the yield significantly. Identify the PGR:

1.	ABA	2.	Gibberellin
3.	Auxin	4.	Ethylene

176. Consider the given two statements:

Assertion (A):	Plant growth is intimately linked to the water status of the plant.
Reason (R):	The plant cells grow in size by cell enlargement, which in turn requires water.

1.	**(A)** is True but **(R)** is False
2.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
3.	Both **(A)** and **(R)** are True and but **(R)** does not correctly explain **(A)**
4.	**(A)** is False but **(R)** is True

177. Consider the given two statements:

Assertion (A):	It is better to consider the respiratory pathway as an amphibolic pathway rather than only as a catabolic one.
Reason (R):	Breaking down processes within the living organism is catabolism, and synthesis is anabolism.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explains **(A)**
3.	**(A)** is True but **(R)** is False
4.	**(A)** is False but **(R)** is True

178. Consider the given two statements:

Assertion (A):	The theoretical maximum yield of ATP per molecule of glucose in aerobic respiration is 38 ATP.
Reason (R):	Each NADH molecule produces 2.5 ATP during oxidative phosphorylation

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**
3.	**(A)** is True but **(R)** is False
4.	Both **(A)** and **(R)** are False

179. The number of times the Krebs cycle turns per glucose molecule will be:

1.	1	2.	2
3.	3	4.	6

180. Which region of the brain contains centers responsible for controlling involuntary reflexes like respiration and cardiovascular functions?
1. Hypothalamus
2. Cerebellum
3. Pons
4. Medulla oblongata

Mitoprep

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